

Triangles, radians and sectors - Set 1 - Solutions

Specification MM4.1, MM4.2, M5.16.

Answer key

Question	Answer	Topic
1	F	Triangles, radians and sectors
2	B	Triangles, radians and sectors
3	C	Triangles, radians and sectors
4	C	Triangles, radians and sectors
5	C	Triangles, radians and sectors
6	C	Triangles, radians and sectors
7	A	Triangles, radians and sectors
8	D	Triangles, radians and sectors
9	E	Triangles, radians and sectors
10	C	Triangles, radians and sectors
11	D	Triangles, radians and sectors
12	A	Triangles, radians and sectors

Worked solutions

Question 1 TRI-01 · Triangles, radians and sectors · spec MM4.1

In triangle ABC , $AB = 5$, $AC = 8$ and angle $BAC = 60^\circ$. Find BC .

Answer: F

Cosine rule: $BC^2 = 25 + 64 - 2 \cdot 5 \cdot 8 \cdot \frac{1}{2} = 89 - 40 = 49$, so $BC = 7$.

Question 2 TRI-02 · Triangles, radians and sectors · spec MM4.1

A triangle has sides 4, 5 and 6. Find the cosine of its largest angle.

Answer: B

The largest angle is opposite the side 6: $\cos \theta = \frac{16 + 25 - 36}{2 \cdot 4 \cdot 5} = \frac{5}{40} = \frac{1}{8}$.

Question 3 TRI-03 · Triangles, radians and sectors · spec MM4.2

A sector of a circle of radius 6 cm has area $15\pi \text{ cm}^2$. Find the arc length of the sector.

Answer: C

$\frac{1}{2}r^2\theta = 18\theta = 15\pi \Rightarrow \theta = \frac{5\pi}{6}$. Arc = $r\theta = 6 \cdot \frac{5\pi}{6} = 5\pi$.

Question 4 TRI-05 · Triangles, radians and sectors · spec MM4.2

Convert 135° to radians.

Answer: C

$135 = \frac{3}{4} \times 180$, so $\frac{3\pi}{4}$.

Question 5 TRI-09 · Triangles, radians and sectors · spec MM4.1

Two sides of a triangle are 3 and 5 and the included angle is θ , where $\cos \theta = -\frac{1}{5}$. Find the length of the third side.

Answer: C

$c^2 = 9 + 25 - 2 \cdot 3 \cdot 5 \cdot \left(-\frac{1}{5}\right) = 34 + 6 = 40$.

Question 6 TRI-13 · Triangles, radians and sectors · spec MM4.2

The minute hand of a clock is 12 cm long. How far does its tip travel in 25 minutes?

Answer: C

25 minutes is $\frac{25}{60} = \frac{5}{12}$ of a revolution: $\frac{5}{12} \times 2\pi \times 12 = 10\pi \text{ cm}$.

Question 7 X2-15 · Triangles, radians and sectors · spec MM4.1

In triangle ABC , P : ' $a^2 = b^2 + c^2$ '. Q : 'angle $A = 90^\circ$ '. Which one of the following is true?

Answer: A

Pythagoras and its converse: the cosine rule gives $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$, which is 0 exactly when $a^2 = b^2 + c^2$.

Question 8 TRI-04 · Triangles, radians and sectors · spec MM4.1

Triangle PQR has area $10\sqrt{3}$, $PQ = 8$ and $PR = 5$. Given that angle QPR is obtuse, find QR .

Answer: D

$$\frac{1}{2} \cdot 8 \cdot 5 \sin P = 10\sqrt{3} \Rightarrow \sin P = \frac{\sqrt{3}}{2}, \text{ and } P \text{ obtuse gives } P = 120^\circ. QR^2 = 64 + 25 - 80 \cos 120^\circ = 89 + 40 = 129.$$

Question 9 TRI-06 · Triangles, radians and sectors · spec MM4.2

A chord of a circle of radius 4 subtends an angle of $\frac{\pi}{3}$ at the centre. Find the area of the minor segment cut off by the chord.

Answer: E

$$\text{Sector area } \frac{1}{2} \cdot 16 \cdot \frac{\pi}{3} = \frac{8\pi}{3}. \text{ Triangle area } \frac{1}{2} \cdot 16 \sin \frac{\pi}{3} = 4\sqrt{3}. \text{ Segment } = \frac{8\pi}{3} - 4\sqrt{3}.$$

Question 10 TRI-07 · Triangles, radians and sectors · spec MM4.1

In triangle ABC , $AB = 6$, $BC = 4$ and angle $BAC = 30^\circ$. How many possible triangles are there?

Answer: C

Sine rule: $\frac{\sin C}{6} = \frac{\sin 30^\circ}{4}$, so $\sin C = \frac{3}{4}$. Since $BC = 4 < AB = 6$ and $4 > 6 \sin 30^\circ = 3$, both the acute and obtuse values of C give valid triangles: the ambiguous case, two triangles.

Question 11 TRI-08 · Triangles, radians and sectors · spec MM4.2

The perimeter of a sector of radius r and angle θ radians is 20, and its area is 16. Find the possible values of r .

Answer: D

$$2r + r\theta = 20 \text{ and } \frac{1}{2}r^2\theta = 16. \text{ From the first } r\theta = 20 - 2r; \text{ then } \frac{1}{2}r(20 - 2r) = 16 \Rightarrow r^2 - 10r + 16 = 0 \Rightarrow r = 2 \text{ or } 8.$$

Question 12 TRI-10 · Triangles, radians and sectors · spec MM4.1

In triangle ABC , $\sin A : \sin B : \sin C = 3 : 5 : 7$. Find angle C .

Answer: A

$$\text{By the sine rule the sides are in ratio } 3 : 5 : 7. \cos C = \frac{9 + 25 - 49}{2 \cdot 3 \cdot 5} = -\frac{15}{30} = -\frac{1}{2}, \text{ so } C = 120^\circ.$$